

Econometrics: Methods and Applications

Test Exercise 1

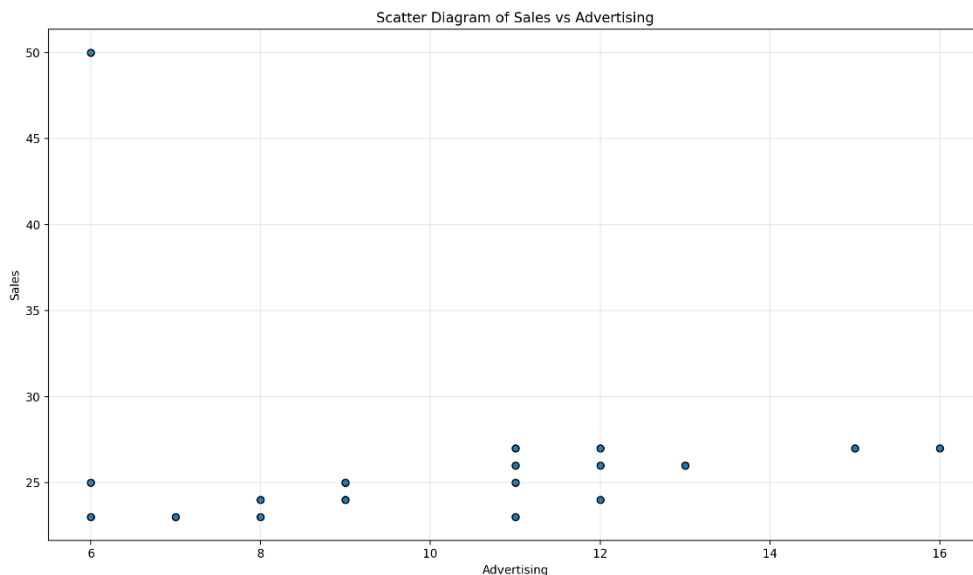
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GitHub repo: [BillDimitro/econometrics-methods-and-applications](https://github.com/BillDimitro/econometrics-methods-and-applications)

Note: This PDF presents the analytical solution, including the formulas, handwritten derivations, final numerical results, plots, and interpretation for Test Exercise 1.

The numerical computations and figures were implemented in Python. The relevant script for this exercise is `Module_2/scripts/test_ex_1.py`, which is included in the GitHub repository above. This repository also contains the training exercises completed during this module.

a) Below you can find the scatter diagram with sales on the vertical axis and advertising on the horizontal axis:



The scatter diagram shows one clear outlier. For most weeks, sales are between 23 and 27, while week nr. 12 has sales equal to 50. Therefore, if we fit a regression line to all the data, this outlier will strongly affect the estimated regression line. The model will probably have a relatively low R^2 , a large residual standard error, and one residual that is much larger than the others.

b) Simple Regression Model $\rightarrow \boxed{S_i = \alpha + \beta A_i + \varepsilon_i}$

- $S_i \rightarrow$ Sales in week i , dependent variable
- $A_i \rightarrow$ Advertising in week i , explanatory variable
- $\alpha \rightarrow$ intercept
- $\beta \rightarrow$ slope coefficient
- $\varepsilon_i \rightarrow$ residual part not explained by advertising

Estimation of coefficients a and b:

$$b = \frac{\sum_{i=1}^n (A_i - \bar{A})(S_i - \bar{S})}{\sum_{i=1}^n (A_i - \bar{A})^2} = \frac{\text{cov}(A_i, S_i)}{\text{var}(A_i)} = -0.325$$

$$a = \bar{S} - b \cdot \bar{A} = 29.627$$

Out of curiosity and to better understand our example, we also calculate the R^2 metric and the residual standard error:

$$R^2 = \frac{b^2 \cdot \text{var}(A_i)}{\text{var}(S_i)} = 0.027$$

$$S = \sqrt{\frac{1}{n-2} \cdot \sum_{i=1}^n e_i^2} = 5.836$$

The very low $R^2=0.027$ means that advertising explains only about 2.7% of the variation in sales.

The residual error is also very large, probably because of the extreme outlier.

The standard error of b is calculated as:

$$S_b = \sqrt{\frac{s^2}{\sum_{i=1}^n (A_i - \bar{A})^2}} = 0.459, \text{ with } s^2 = \frac{1}{n-2} \sum_{i=1}^n e_i^2$$

The t -value of b is calculated as:

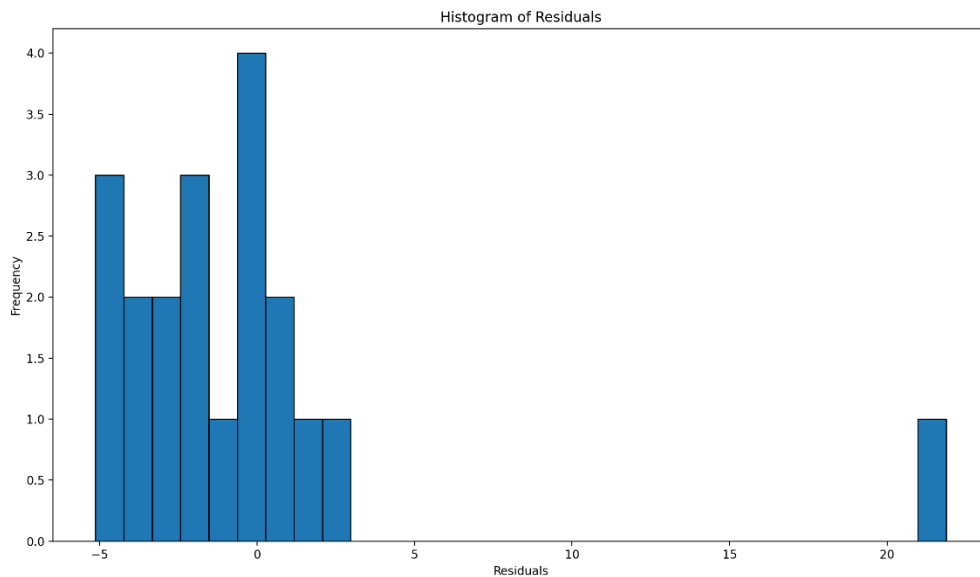
$$t_b = \frac{b}{S_b} = -0.707, \quad |t_b| = 0.707$$

Since the absolute value of the t -statistic is much smaller than the 5% critical value (≈ 2), b is not significantly different from zero.

Therefore, based on this regression using all 20 observations, advertising does not appear to have a statistically significant effect on sales. This result is probably distorted by the extreme outlier observed in the scatter plot.

Note: The parameter calculations were implemented in Python using the formulas above. The corresponding code is included in the provided GitHub repository.

c) Below you can find the histogram of the calculated residuals:



The histogram shows one residual that is much larger than all the others. Most residuals are concentrated around zero, approximately between -4.7 and 2.6, but one residual is 22.3. Therefore, the residuals are not symmetrically distributed, which means that assumptions A3 and A4 are violated.

d) Since the large residual corresponds to the special week in which the store was also open during the evenings, the high sales this week are not caused only by advertising, but also by the longer opening hours. Therefore, I would remove this datapoint from the sample and estimate the regression again, using the remaining 19 observations.

e) After deleting the outlier week from the sample and using the remaining 19 weeks to estimate the regression again, the results are:

$$b = 0.375, \quad s_b = 0.088, \quad t_b = 4.252$$

$$\alpha = 21.125, \quad s = 1.054, \quad R^2 = 0.515$$

The value of the t -statistic is now much larger than the 5% critical value (± 2), hence b is significantly different from zero. Therefore, after removing the special week, advertising appears to have a statistically significant positive effect on sales.

f) Comparing parts (b) and (e), we see that the special week had a very large effect on the regression. When removing the outlier, the value of R -squared increased from 2.7% to 51.5%, and the residual standard error fell from 5.836 to 1.054.

Therefore, the model fits the remaining data much better.

The lesson here is that, even a single unusual observation, can strongly distort a regression result, violating the regression assumptions, and creating a misleading estimate of the advertising effect.